Medicare Provider Fraud Risk Scoring System

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GitHub repository: https://github.com/atto18/Insurance_Fraud_ML
DockerHub repository: https://hub.docker.com/r/marcg7/ml-insurance-fraud

1. Project Overview

This project aims to build a machine learning system that assigns a fraud-risk score to healthcare providers using Medicare claims data and exclusion-based labels. The objective is not to detect proven fraud, but to identify providers with suspicious patterns based on historical data and known exclusions.

2. Data Sources and Preparation

Two main datasets were used:

- **Provider Dataset**: Medicare provider-level billing and utilization data
- **Exclusion Dataset**: OIG list of excluded individuals / entities

Data Processing Steps:

1. Raw datasets were cleaned and preprocessed
2. Datasets were joined using `provider_id` / NPI
3. A binary label was created: 1 → fraud, 0 → no fraud
4. Final labeled dataset was generated
5. Feature engineering was applied to the label dataset to create additional variables capturing financial ratios, utilization patterns, and anomaly indicators. This is the final dataset used.

3. Modeling Approaches

3.1 - Method 1: Standard Training (Full Dataset):

Four models (*Logistic Regression*, *LightGBM*, *XGBoost*, *CatBoost*) were trained on the full dataset while addressing class imbalance. Their performance was evaluated using `AUROC`, `AUPRC`, and `Precision@K`. The final prediction was obtained using an ensemble method that combines the outputs of all models, allowing the system to leverage their complementary strengths and produce more accurate and stable fraud scores.

3.2 - Method 2: Iterative Training (Balanced Sampling):

An iterative ensemble using LightGBM and XGBoost was used to handle class imbalance. Each iteration trained on all 187 positive samples and 200 randomly sampled negatives, repeated over 100 iterations. Final predictions were averaged, improving detection of rare fraud cases.

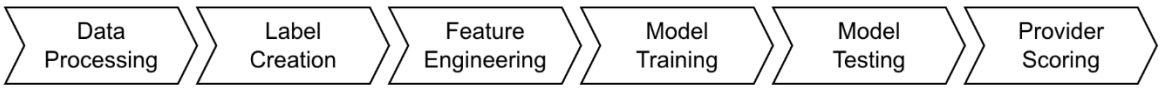
4. Results and Evaluation

Due to the strong class imbalance, AUPRC is the most important metric as it measures the model's ability to correctly detect fraud cases, while AUROC evaluates the overall ranking between fraud and non-fraud providers. The iterative method achieved a higher AUROC, indicating better ranking performance. However, the original ensemble achieved a significantly higher AUPRC making it more effective at detecting rare fraud cases. Therefore, the original ensemble is better suited for this problem.

Model	AUROC	AUPRC	P@50	P@100	P@200	P@500
LR	0.7044	0.0008	0.0000	0.0000	0.0050	0.0020
LightGBM	0.6907	0.0003	0.0000	0.0000	0.0000	0.0000
XGBoost	0.7747	0.0020	0.0000	0.0200	0.0100	0.0040
CatBoost	0.7972	0.0018	0.0200	0.0100	0.0100	0.0060
Ensemble	0.7952	0.0148	0.0200	0.0100	0.0050	0.0040

Model	AUROC	AUPRC	P@50	P@100	P@200	P@500
Iterative Ensemble	0.8124	0.0013	0.0000	0.0000	0.0050	0.0040

5. System Pipeline



6. Conclusion

This project shows how machine learning can identify high-risk providers despite imbalanced data. The model provides useful risk scores, but results should be seen as indicators, not proof of fraud.